

Chapter 3. GROUPS

3.3 Cyclic Groups

3.3.1 Some important properties of Cyclic Groups



Today we will study **some IMPORTANT properties of cyclic groups**, like, for example, they are all abelian, and all its subgroups are cyclic.

3.3.1 Some IMPORTANT properties of cyclic groups

A cyclic group can have more than one generator:

$$\mathbb{Z}_{12} = \{\bar{0}, \bar{1}, \bar{2}, \dots, \bar{11}\}$$

We can easily prove that $\bar{5}$ also generates $\mathbb{Z}_{12}$

$$\mathbb{Z}_{12} = \langle \bar{5} \rangle = \{k\bar{5} \mid k \in \mathbb{Z}\}.$$

$$0 \cdot \bar{5} = \bar{0}$$

$$1 \cdot \bar{5} = \bar{5}$$

$$2 \cdot \bar{5} = \bar{5} + \bar{5} = \bar{10}$$

$$3 \cdot \bar{5} = \bar{5} + \bar{5} + \bar{5} = \bar{3 \cdot 5} = \bar{15} = \bar{3}$$

$$15 = 1 \cdot 12 + \underline{3}$$

How can we know that a given element is a generator?

Can we find all of the generators?

THEOREM 3.28. Let $G = \langle g \rangle$ be a cyclic group with generator g . Then:

- (1) G is abelian.
- (2) Every subgroup of G is cyclic.
- (3) Suppose that $o(g) = n$. Then $G = \langle g^k \rangle$ if and only if $\gcd(k, n) = 1$.

Proof.

(1) If $x, y \in G$, then $x = g^k, y = g^m$ for some integers $k, m \in \mathbb{Z}$. Using the Exponent laws we obtain that

$$\underline{xy} = g^k g^m = g^{k+m} = g^{m+k} = g^m g^k = \underline{yx}$$

This proves that G is abelian.

(2) Take H a subgroup of G . If $H = \{1\}$ then we are done because $H = \langle 1 \rangle$ is cyclic. Suppose that $g^k \in H$ (with $k \neq 0$). We can assume that $k > 0$; if $k < 0$, then

$-k > 0$ and $g^{-k} = (g^k)^{-1} \in H$ (H is a subgroup). By the Well-Ordering Principle we can choose m to be the smallest positive integer such that

$$\underline{g^m \in H.}$$

Claim. $H \stackrel{?}{=} \langle g^m \rangle$

Notice that $\langle g^m \rangle \subseteq H$ because $g^m \in H$ and H is a subgroup. It remains to show that $H \stackrel{?}{\subseteq} \langle g^m \rangle$.

Take $g^s \in H$ and apply the Division Algorithm to write

$$s = qm + r, \text{ with } 0 \leq r < m.$$

From here, we have $r = s - qm$ and

$$\underline{g^r} = \underbrace{g^s}_H \cdot \underbrace{(g^m)^{-q}}_H \in \underline{H}.$$

$\uparrow$
 H is a subgroup

But $r < m$, and m was the smallest positive integer satisfying that $g^m \in H$.

So, $r = 0$ and $\underline{s = qm}$. Thus:

$$g^s = g^{qm} = (g^m)^q \in \langle g^m \rangle.$$

From here, we obtain that $H = \langle g^m \rangle$,
that is, H is cyclic.

(3) Suppose now that $\boxed{o(g) = n}$.

$$\left. \begin{array}{l} G = \langle g^k \rangle \\ g^k \text{ is a generator} \\ \text{of } G \end{array} \right\} \stackrel{?}{\iff} \gcd(k, n) = 1$$

$\Rightarrow$ | Suppose first that $G = \langle g^k \rangle$.

In particular, $g \in G = \langle g^k \rangle$. So,

$$g^1 = (g^k)^m = \underline{g^{km}} \text{ for some integer } m.$$

From Theorem 4.5 we obtain that

$$1 \equiv km \pmod{n}.$$

This means that $n \mid (1 - km)$. Then

$$1 - km = qn, \text{ for some integer } q.$$

That is to say that

$$1 = km + gn, \text{ for integers } m \text{ and } g.$$

From here we obtain that

$$\underline{\gcd(k, n) = 1}.$$

$\Leftarrow$ | Suppose now that $\gcd(k, n) = 1$.

We need to show that $G = \langle g^k \rangle$.

Since $G = \langle g \rangle$, it is enough to show that $g \in \langle g^k \rangle$.

Bézout's lemma tells us that

$$\underline{1 = xk + yn}, \text{ for some integers } x \text{ and } y.$$

Using the Exponent laws we obtain that

$$\begin{aligned} g &= g^1 = g^{xk+yn} = g^{xk} \cdot g^{yn} = \\ &= (g^k)^x \cdot (g^n)^y \underset{g^n=1}{=} (g^k)^x \in \langle g^k \rangle \end{aligned}$$

Remarks. ① Every cyclic group is abelian.
The CONVERSE is NOT true: There
are abelian groups that are not cyclic.
For example: $\mathbb{Z}_8^*$ is abelian
but not cyclic.

$$\textcircled{2} \quad \mathbb{Z}_{12} = \{\bar{0}, \bar{1}, \bar{2}, \dots, \bar{11}\}$$

The generators of $\mathbb{Z}_{12}$ are the elements $\bar{k}$ such that

$$\gcd(k, 12) = 1.$$

Generators of $\mathbb{Z}_{12}$: $\bar{1}, \bar{5}, \bar{7}, \bar{11}$
